

BCS-031

1. What are virtual functions and what are its uses ? Explain with example.
What are Pure virtual functions and what are its uses ? Explain with example.
2. What is a friend function ? How does it differ from the member function ? Explain the implementation of friend function in C++, with the help of an example program.
3. Differentiate between member functions and global functions with help of an example.
4. What is an inline function ? How does the execution of inline functions differ from normal functions ? Explain the advantages of inline function, with suitable example
5. What is object oriented programming ? Explain the benefits of object oriented programming. How does object oriented programming differ from structured programming ?
6. Define the term class. How are class and object related ? Explain with example.
7. Compare Class templates and Function templates with the help of example code.
8. What are constructors ? Write the characteristics of a constructor. What are the limitations of a constructor ? What is a Destructor in C++ ? Do constructors and destructors have return type ?
9. Differentiate between copy constructor and default constructor in C++, with the help of an example for each
10. Short note on: Default and Parameterized Constructor. New and Delete Operator
11. What is STL ? Explain the importance of STL library in C++.
12. What is Type Conversion ? What is the advantage of Type Conversion ? Briefly discuss Type Casting and Automatic Type Conversion.
13. What is Polymorphism ? What are the advantages of polymorphism ? Mention the types of polymorphism supported by C++.
14. What is Encapsulation ? Explain the difference between information hiding and encapsulation with the help of an example.
15. What is inheritance ? Briefly discuss the different types of inheritance, supported by C++. Explain how inheritance is implemented in C++.
16. What is an access specifier ? Explain different types of access specifiers available in C++.
17. Differentiate between early binding and late binding with an example of each.
18. What are Stream Manipulators ? Briefly discuss the purpose of various stream manipulators.
19. What is message passing ? Explain with the help of an example.
20. What is 'this' pointer ? Explain the significance of 'this' pointer with the help of an example program.
21. What is the purpose of exception handling ? Explain how exception is handled in C++, with the help of an example.
22. Explain the difference between method overloading and method overriding with example.
23. What is the difference between function overloading and function overriding in C++ ? Explain the usage of these concepts, with suitable example in C++.
24. What is operator overloading ? Write a C++ program to overload '+' operator to find the sum of two complex numbers. Support your program with suitable comments.